Quick guide to carbonate rock descriptions and interpretation

I. Carbonate grains

I.a Non-skeletal carbonate grains

Guide for grain identification

Hard skeleton of organisms…….....Skeletal grain (see I.b)

A) Individual grains

Components without internal structure……..1
Components with internal structure……....2 and 3

1) a) on average spherical, elliptical, rounded; composed of micrite; commonly in clusters; size often <0.2mm…….Peloid
1) b) highly variable shape, rounded or angular edges; composed of micrite; size commonly >0.2mm……...Intraclast

2) a) regular concentric layers around nucleus
  2) a) 1 regular shapes, typically spherical, elliptical; partly radial structure; size <2mm……Ooid
  2) a) 2 irregular shapes; very regular thin concentric micritic layers; nuclei of rock fragments, pisoids; size typically 2mm…….Pisoid
2) b) irregular layers all around
  2) b) 1 on average irregular elliptical, spherical grains; internal micritic layers not concentric, rather overlapping; biogenic accretion around nucleus; size mm/cm……...Oncoid
  2) b) 2 rounded bioclasts/lithoclasts/peloids with thin micritic rim all around; boundary 'core' and 'rim' gradual; sometimes evidence for small burrows (size 5-10 μm) with/without micrite……Cortoid

B) Composite grains

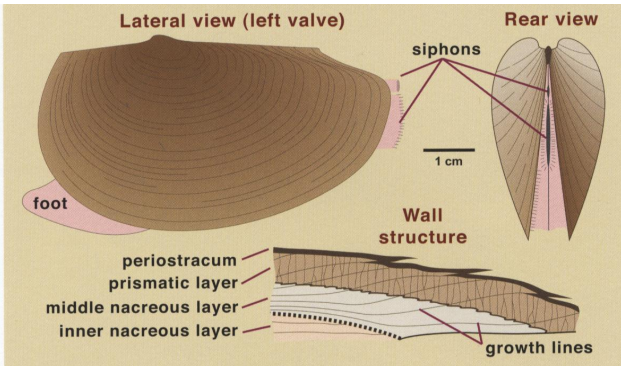
3) a) irregular aggregates of several component grains, outline with several lobes/embayments; ooids, bioclasts, peloids cemented together by sparite/micrite or algal encrustrations; size mm/cm…Aggregate grain
3) b) variably shaped, partly rounded micritic grains with 'inclusions' of other components……Intraclast
3) c) carbonate/non-carbonate rock fragments; angular to very well rounded; internal fossils 'cut off'; no compaction; polymict…..….Extraclast

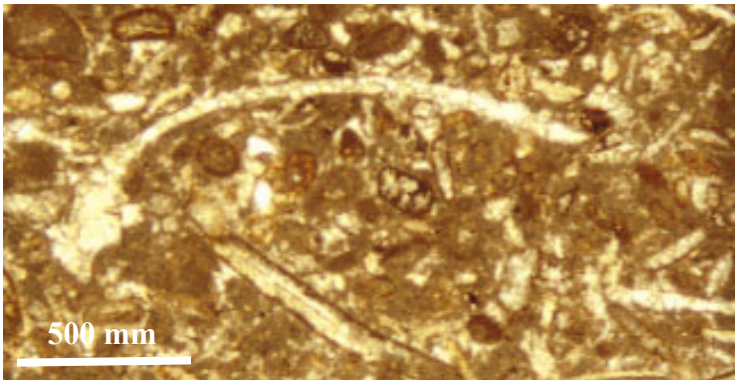
PELOIDS		Small micritic grains, commonly without internal structure. Subrounded, spherical, ovoid or irregular in shape. Size between <0.02 and about 1 mm, commonly 0.10 to 0.50 mm.
COATED GRAINS	CORTOIDS	Rounded skeletal grains and other grains covered by a thin micrite envelope. Boundary between the central grain and the envelope indistinct. Size between <1 mm to a few centimeters.
	ONCOIDS	Large and small grains consisting of a more or less distinct nucleus (e.g. a fossil) and a thick cortex formed by irregular, non-concentric, partially overlapping micritic laminae. Laminae may exhibit biogenic structures. No tendency to increase sphericity during growth. Size from <1 mm to a few decimeters.
	OOIDS	Spherical or ovoid grains, consisting of smooth and regular laminae formed as successive concentric coatings around a nucleus. Laminae may exhibit tangential and radial microfabrics. Size between 0.20 and about 2 mm, commonly between 0.5 and 1 mm.
	PISOIDS	Large subspherical and irregularly shaped grains, consisting of a mostly non-biogenic nucleus and a thick cortex formed by conspicuously, often densely spaced laminae exhibiting tangential and radial microfabrics. Pisoids occur as isolated grains or are incorporated in crusts. Size generally >2 mm, up to >1 cm.
GRAIN AGGREGATES		Compound grains consisting of two or more originally separated particles (e.g. ooids, skeletal grains) that have been bound and cemented together, forming grape-like or rounded lumps. Intergrain spaces filled with micrite or spar. Outline irregular lobular or rounded. Size 0.5 to more than 2 mm.
CLASTS		Synsedimentary or postsedimentary lime clasts, reworked partly consolidated carbonate sediment or already lithified material. Shape and size are highly variable: angular to rounded. Size ranges between <0.2 mm and several decameters. Very small clasts are hardly distinguishable from peloids.
SKELETAL GRAINS		Fragmented or complete skeletons of organisms. Size from 0.05 mm to many centimeters.

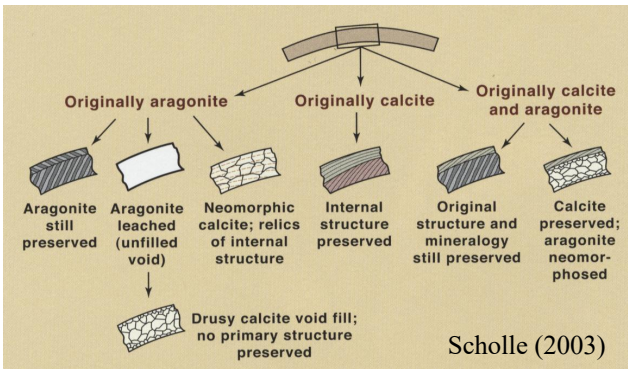
Flügel (2004)

I.b Typical fossils in thin section (bioclasts)

1) Bivalves



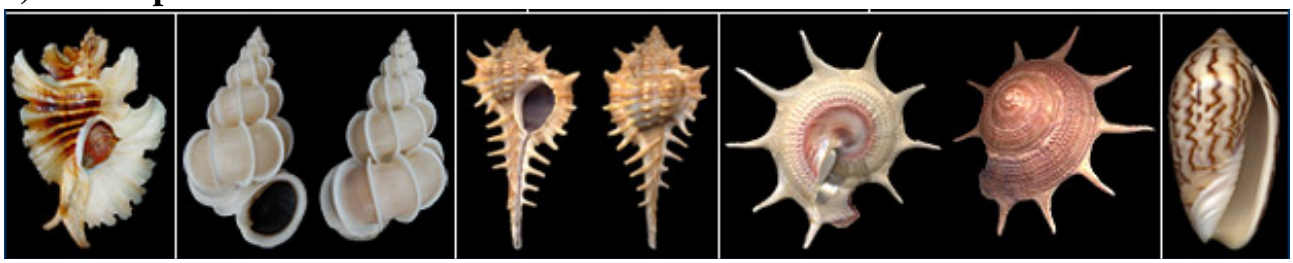


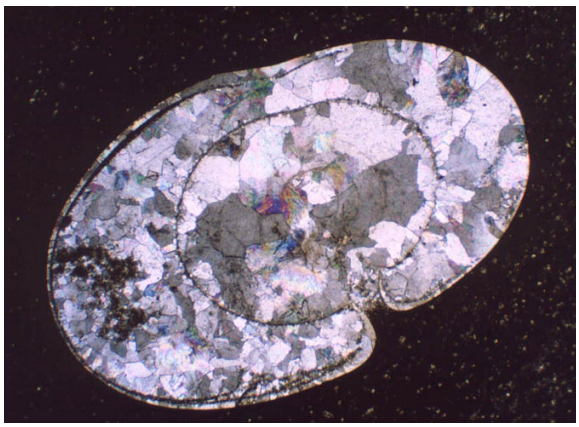


Scholle (2003)

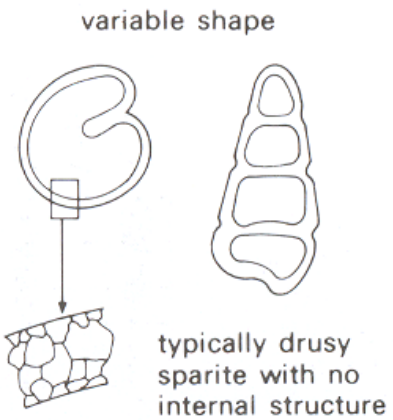
- Two symmetrical valves (in contrast to brachiopods)
- Usually found disarticulated
- Elongate, rectangular to curved grains in thin section
- Original mineralogy determines preservation

2) Gastropods





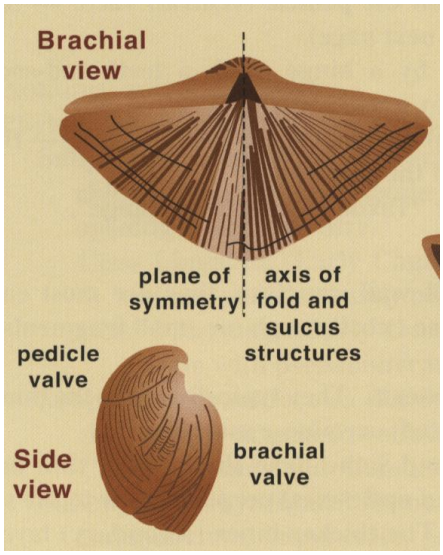
Gastropod filled with calcite cement in mud (10mm) Corlett (2011)

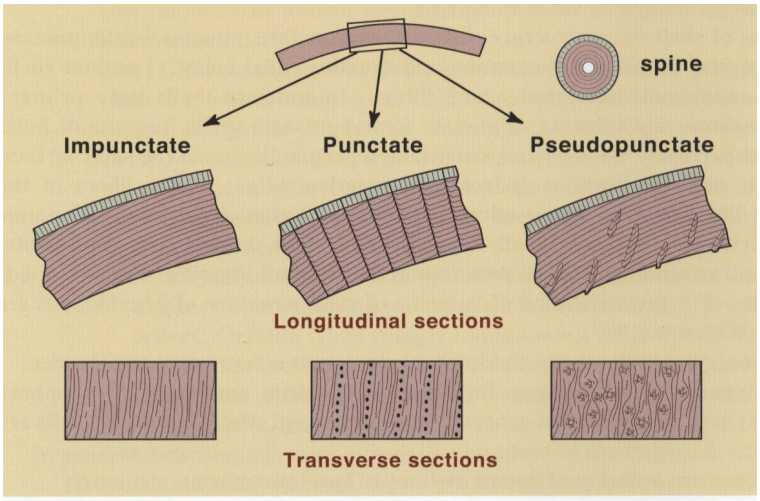


Tucker (2001)

- Characteristic spiral shape
- Usually aragonite: Mold, or preserved as drusy calcite

3) Brachiopods

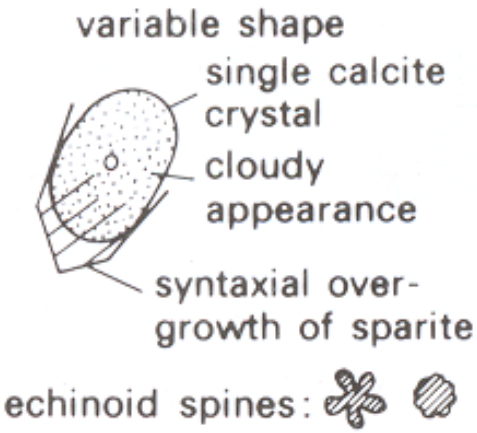




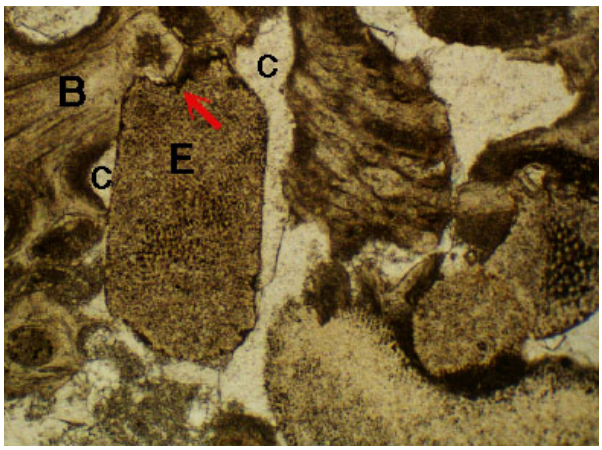
Scholle (2003)

- Two asymmetrical valves (in contrast to bivalves)
- Usually found articulated
- low-Mg calcite: internal structure usually preserved

4) Echinoderms

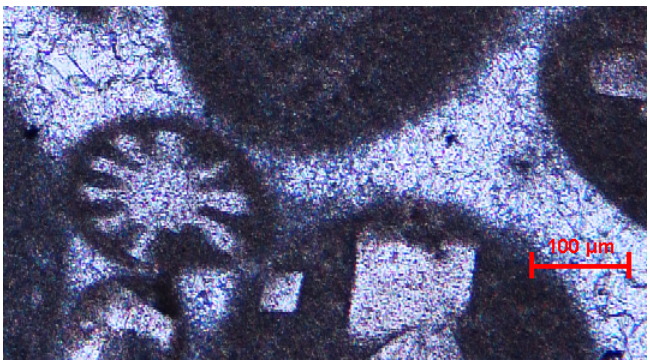


Tucker (2001)



Corlett (2011)

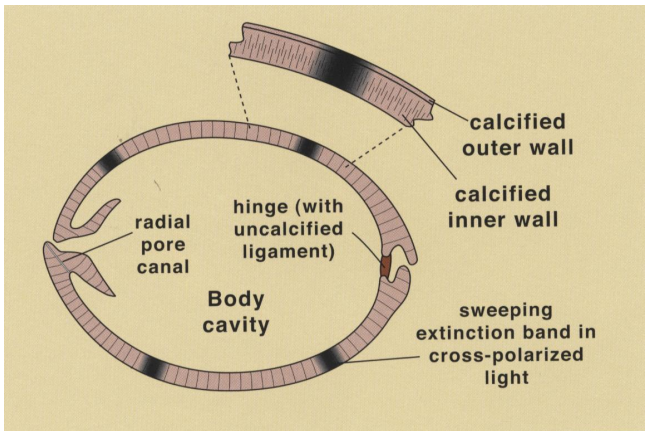


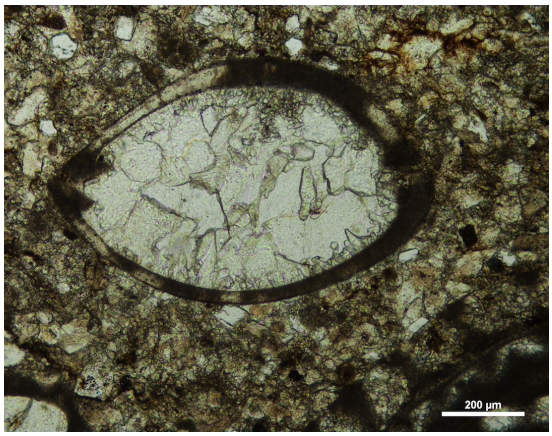


- Calcite, mottled or spotted appearance
- Characteristic uniform extinction and syntaxial overgrowth cements

5) Arthropods

Ostracod



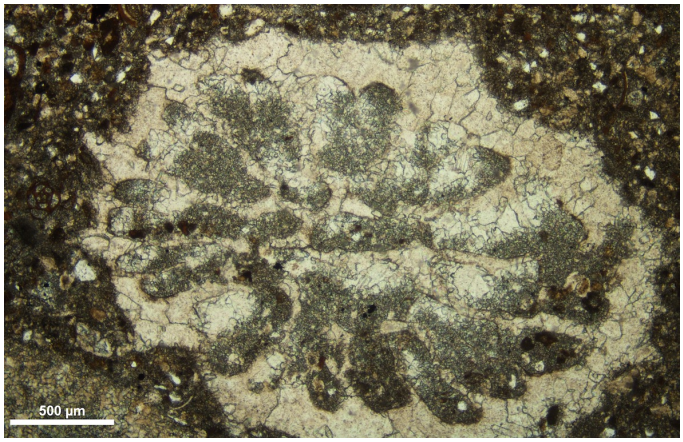


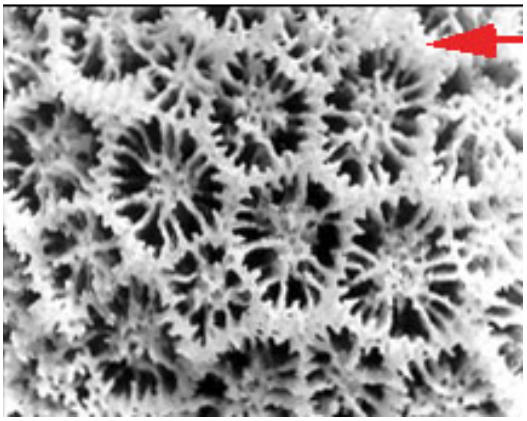
Scholle (2003)

• Ostracods: calcite with radial-fibrous structure (undulose extinction)

6) Cnidaria (corals)

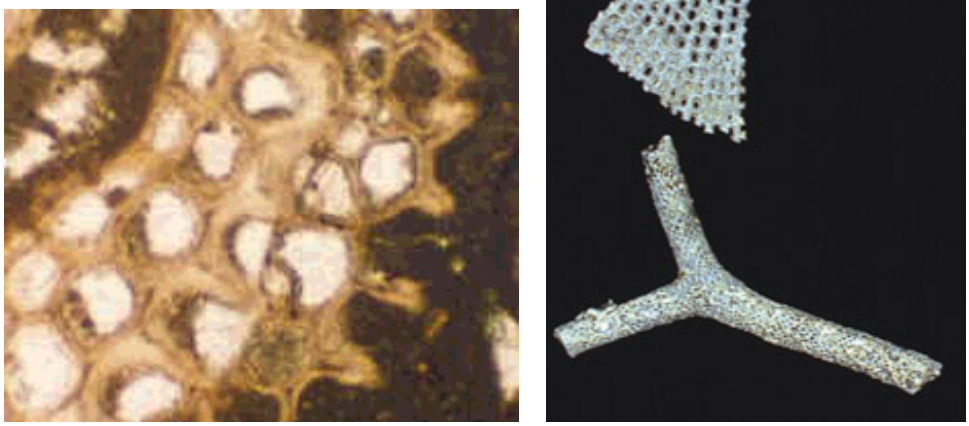
• Paleozoic rugose and tabulate corals: high-Mg calcite, good preservation
• Scleractinian corals: aragonite, poor preservation (commonly dissolved or replaced)





http://www.biophysics.sbg.ac.at

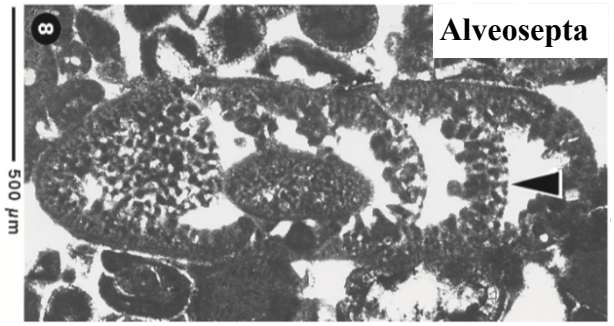
7) Bryozoa



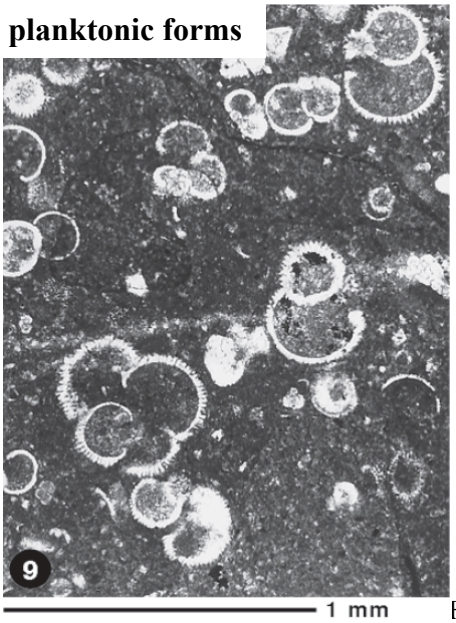
Corlett (2011)

- Aragonite, calcite, or both
- Recognized by fenestral pores, smaller than in corals

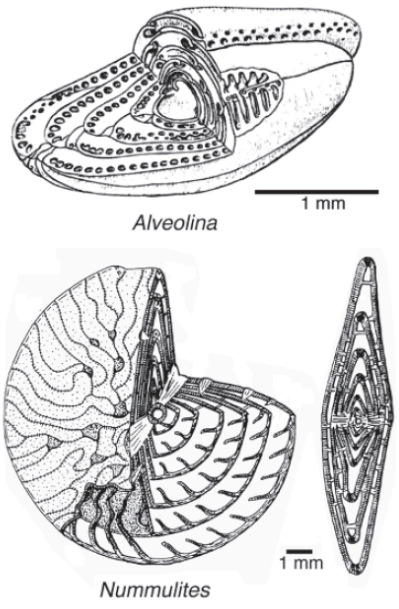
8) Foraminifera



- Diverse shape:
  - Benthic forms commonly coiled
  - Planktonic forms commonly sub-circular with chambers, "popcorn"
- Dominantly composed of low-Mg or high-Mg calcite



Flügel (2004)



Flügel (2004)

9) Algae (and others)

Red algae

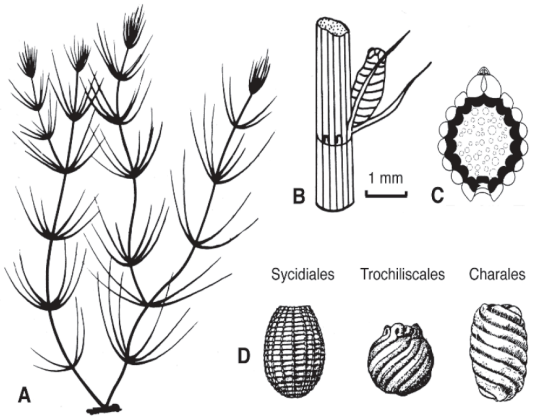


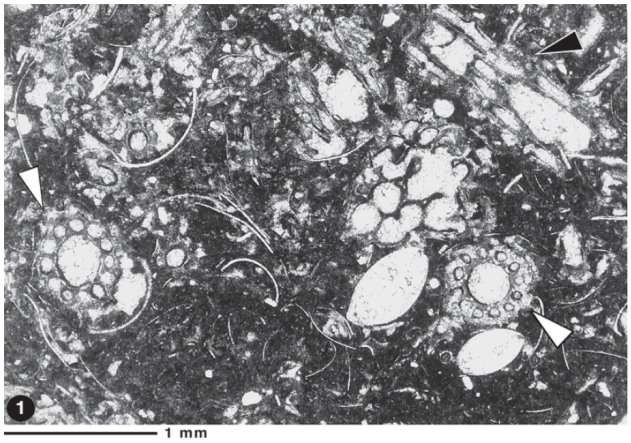
Green algae



Corlett (2011)

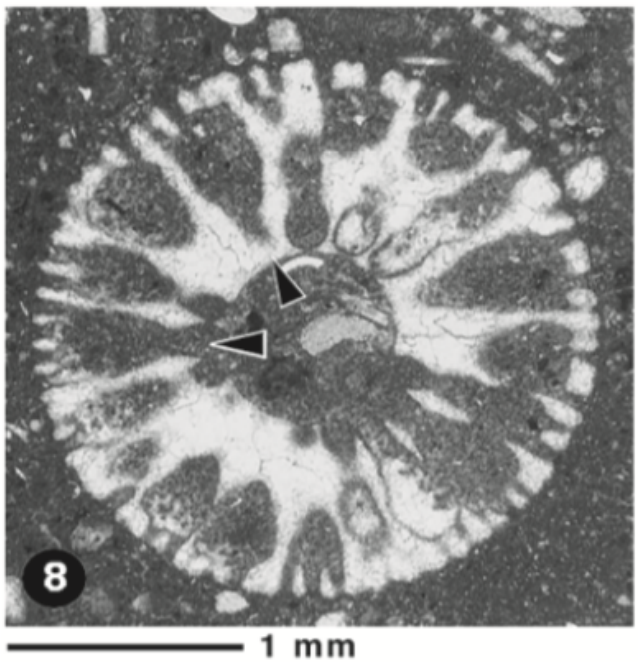
Charophytes (freshwater algae)



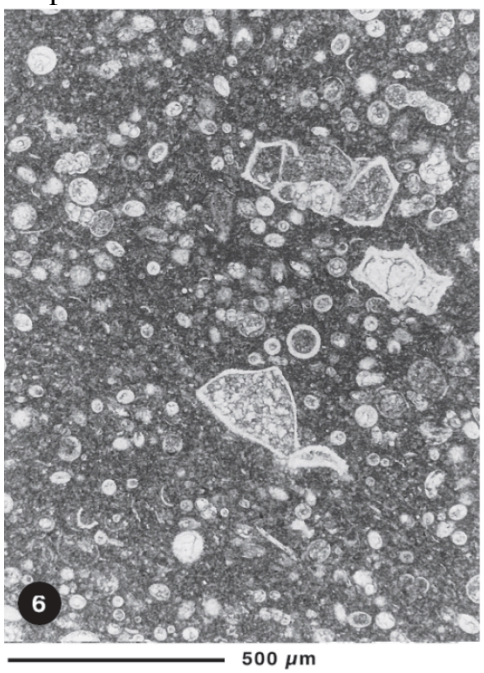


Flügel (2004)

Dasyclads (green algae)



Calcispheres and Globotruncana foraminifera



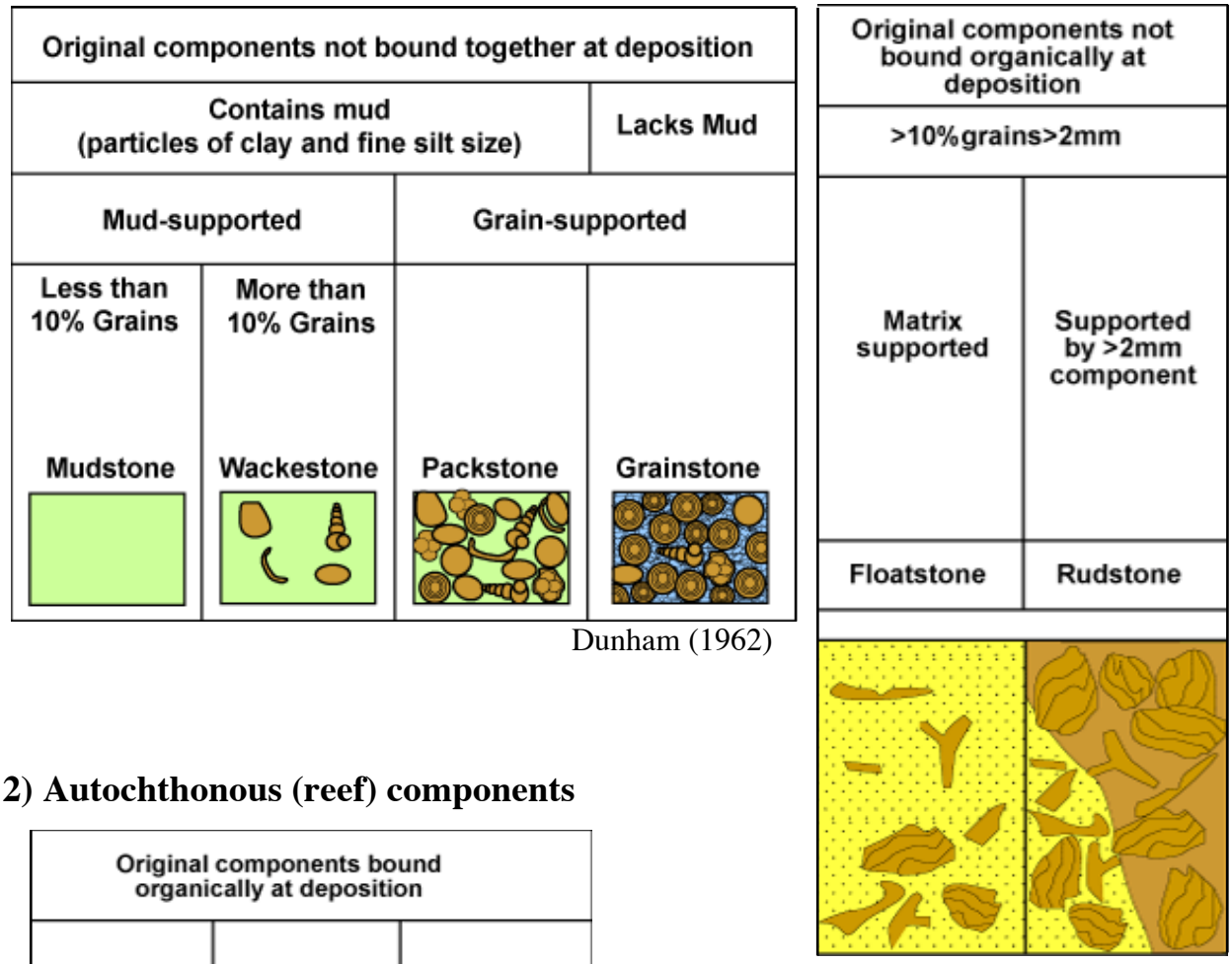
Flügel (2004)

II. Carbonate rock classifications

Dunham classification for carbonate rocks

1) Allochthonous components

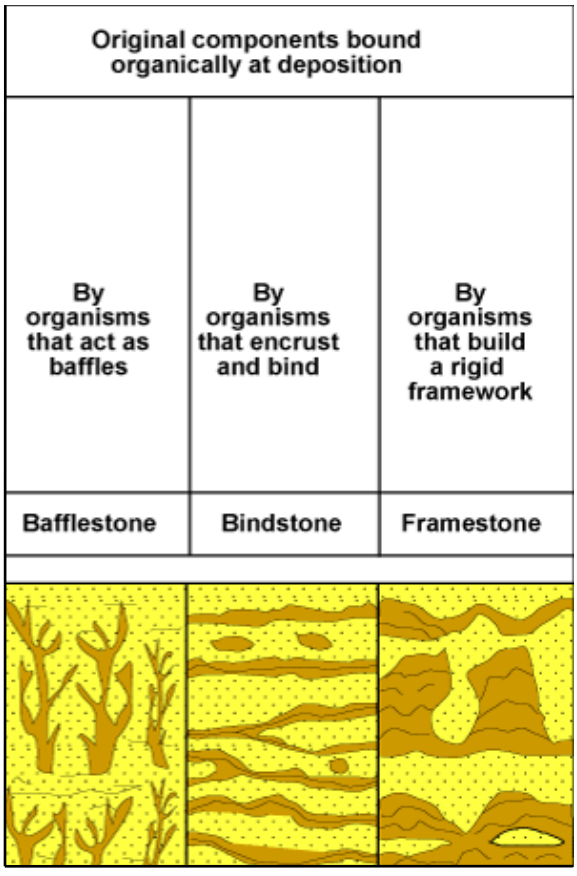
< 2mm allochems **> 2mm allochems**



Dunham (1962)

Modified from Embry and Klovan (1971) and James (1984)

2) Autochthonous (reef) components



Embry and Klovan (1971), James (1984)

III. Sample description

Checklist for carbonate rock descriptions (field/hand sample and thin section)

Color
- both weathered and fresh
- may indicate mineralogy – dolomites commonly yellowish/brownish, red in presence of Fe - may be better visible on weathered surfaces

Lithology
- limestone, dolomite, marl, mixed carbonate-clastic (NB: use adjectives such as 'quartzose', 'shaley'…. when describing carbonate-clastic mixtures, but avoid calcarenite, which should be restricted to grain size)
- quick check with HCl

Bedding and sedimentary structures
- thickness of beds (laminated, thin, thick, massive)
- lateral extent of beds (compare also bioherm/biostrome for organic accumulations)
- bedding surfaces, contacts (sharp, gradational, erosive, wavy, presence of clay seams, diagenetic features such as stylolites…..)
- sedimentary structures in beds (lamination, cross-bedding/-lammination, ripples, graded beds….; includes massive beds without structures – note this can indicate primary absence of structures or secondary destruction due to bioturbation or diagenesis)
- base-of-bed structures: load casts, backfilled channels, gutter casts…..
- any features useful for paleocurrents (ripples, aligned shells in coquinas…)
- very early diagenetic features such as mudcracks, fenestrae, edgewise conglomerates as direct indicators for processes and environments…..
- look for geopetal structures as indicators of stratigraphic up, convex/concave up of shells as indicators of in-situ vs transported nature, life position of fossils such as corals
- cyclicity?

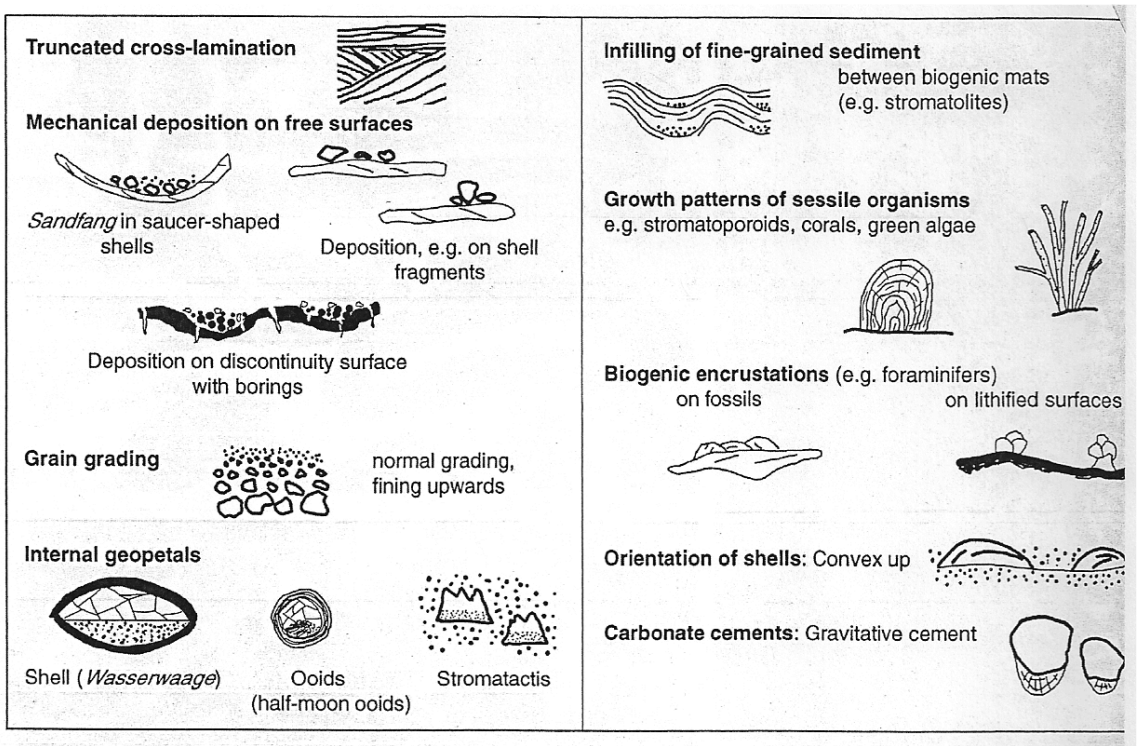
Grain size and texture
- calcilutite, calcisiltite, calcarenite, calcirudite are useful field descriptions
- sorting, grading, roundness, presence/absence of micrite (mud) – grain-/mud-supported texture
- packing of fossils as indicators for transport processes/in-situ accumulation
- note that grain size/shape/sorting in carbonates are controlled primarily by original size and shape of biogenic skeletons, abrasion processes during (limited) transport, but also energy level

Components
- identify and quantify grains, micrite, sparite and porosity if present (use percentage chart or point counting)
- grain types (skeletal/bioclast, non-skeletal), order them according to abundance
- any characteristic faunal associations
- note degree of grain preservation, in particular for bioclasts: entire or broken, original shell preserved or replaced/dissolved (mold)
- micrite: appears pasty in hand sample, fine-grained & dense in thin section; various sedimentary and diagenetic origins - not equivalent to siliciclastic mud
- sparite: appears clear, like glass in hand sample, identifiable crystals/crystal boundaries in thin section; cement and product of diagenetic recrystallization

Biogenic structures
- bioturbation: degree, type

Diagenetic fabrics
- replacement fabrics, dolomitization
- pressure solution (mechanical/chemical)
- nodular fabrics or beds, mineralogy of nodules (silica, Fe….)
- fractures, orientation, width, open/cemented
- type of porosity, estimate amount (none, low, fair, good, excellent; or use percentage chart)

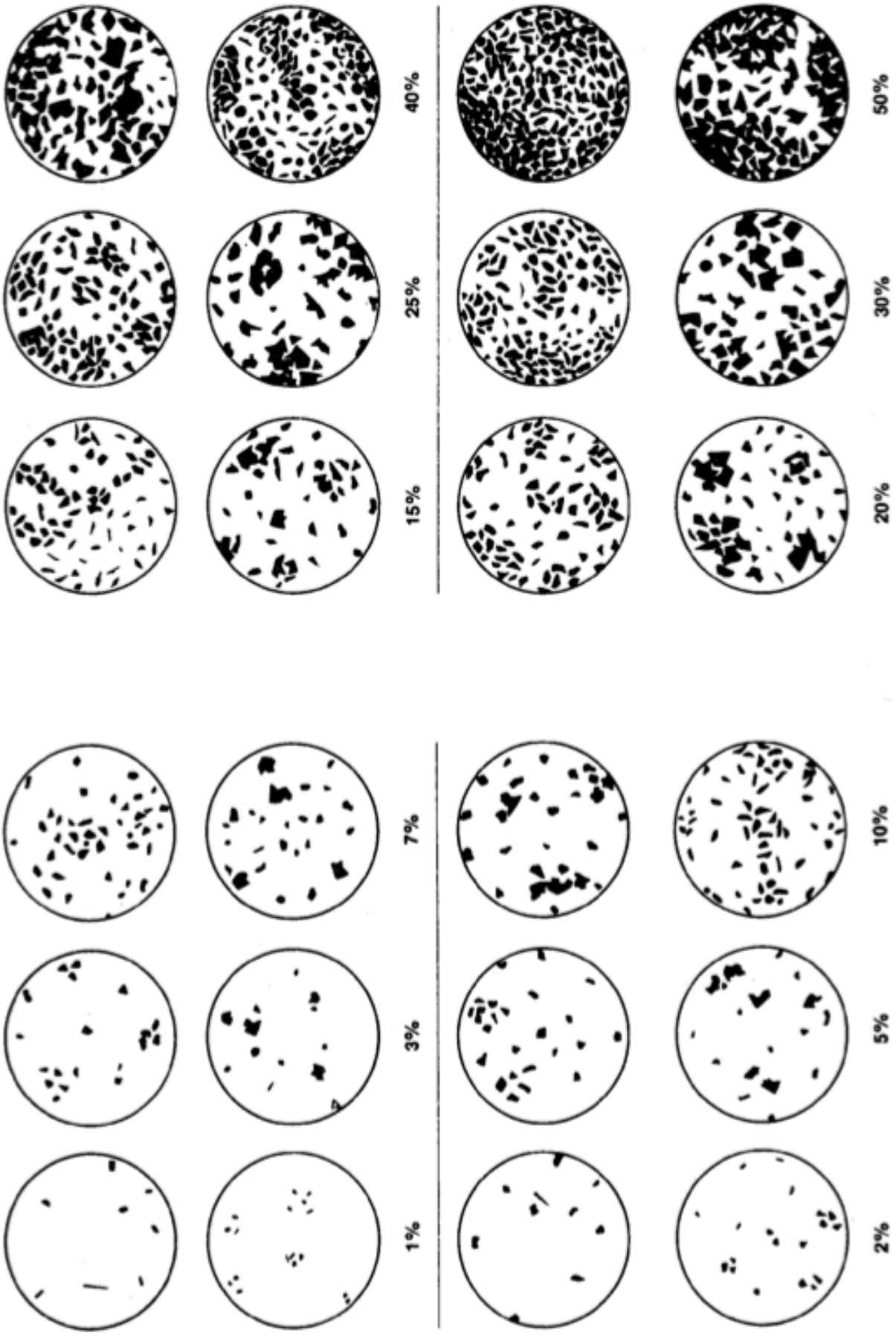
For classification schemes (Dunham in particular)
- consider whether rock derives from activity of benthic sessile organisms (autochthonous – bound together during deposition), or consists of grains, mud and/or cement (allochthonous – not bound together during deposition)
- autochthonous carbonates:
    - sessile organisms form low-relief, tightly packed 'crusts' or accretion structures- *bindstone*
    - sessile organisms form vertical structures separated by sediment – *bafflestone*
    - sessile organisms form framework with intervening incrustations - *framestone*
- allochthonous carbonates: Dunham works on the basis of 3 simple questions: presence/absence of mud?, grain- vs mud-supported fabric?, percentage of grains?
- apply the Embry and Klovan classification for autochthonous carbonates when the role of sessile benthic organisms can be demonstrated; otherwise use *boundstone*
- integrate the dominant or ecologically important fossil group, or dominant grain type into the name, e.g. (branched) coral framestone
- for grainstones, additional clues as to sorting and/or grain size should be included when possible, as this is relevant for porosity evolution
- specific concepts are required for reefal, non-marine and recrystallized carbonates
- and finally - "A classification is simply a tool for organizing information, not the source of a conclusion" (Blatt, 1972)



Common geopetal fabrics in carbonates (Flügel, 2004)



Field criteria for carbonate rocks (black dots: microfacies, white circles: hand samples) (Flügel 2004).



Percentage estimation chart